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Portable Integrated farming with IoT

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Project Guide

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Introduction

Integrated Portable farming refers to innovative, movable agricultural systems designed to maximize income in small or urban spaces, often utilizing technology to allow for year-round, high-density cultivation. These systems, which can be moved to different sites, are particularly suited for rooftop gardens, balconies, or as temporary, sustainable solutions.

Portable farming integrating IoT uses sensors , and automation to create efficient, data-driven micro-farms (like hydroponics, vertical farms) that monitor and control water, nutrients, and environmental conditions remotely, optimizing resource use, boosting yields, and reducing labor for urban, small-scale, or modular food production anywhere.



MOTIVATION

- Unpredictable climate affects traditional agriculture(Due to drought and floods)
- Inefficient use of water and nutrients by conventional irrigation methods
- Increasing population and rapid urbanization results in limited farmland, soil degradation and contamination
- Demand for pesticide-free and safe and healthy food
- Demand for faster crop growth and shorter harvest cycles

CHALLENGES

- **Sensor accuracy & calibration issues** (Temperature, pH, EC sensors) can give wrong readings if not calibrated properly.
- **Data handling complexity:** Large sensor data needs proper storage, cleaning, and analysis.
- **Difficulty in Crop Selection:** Selecting suitable crops is challenging because not all plants adapt well to controlled portable environments.
- **Cost & Maintenance:** High initial investment in sensors and control units, along with regular maintenance of system components, increases overall system complexity.

Problem Statement

Farmers, urban growers, and institutions require a reliable way to produce fresh crops despite limited space, water scarcity, and unpredictable weather, which cause inconsistent yield and quality in open-field farming. This creates the necessity for integrated cultivation that supports growth stability through real-time environmental sensing and adaptive system-level responses.

Objectives

- To conduct survey and study the need for the integrated smart farming system
- To develop a digital twin for analysis of controlled farming conditions.
- To implement a portable IoT-based integrated farming system

Literature Review

S no.	Author	Year	Journal	Topic	Outcome
1	D. C. Love et al.	2015	Aquaculture	Commercial aquaponics production and profitability	Study of 257 aquaponics farms worldwide showed that aquaponics is a promising sustainable food production system combining fish and plant cultivation. Profitability depends on technical knowledge, climate conditions, and diversified revenue sources.
2	Zou et al.	2023	Cell Reports Physical Science	Radiative cooling film for greenhouse cooling	Cooling film reduced greenhouse temperature by 18.6°C, improved crop growth, and reduced water consumption by 64.8%.

S. No	Author	Year	Journal	Topic	Outcome
3	Giacalone, D. & Jaeger, S.R.	2023	Journal of Cleaner Production	Consumer acceptance of novel sustainable food technologies: A multi-country survey	Urban farming and modified-atmosphere vegetables showed highest acceptance; plant-based foods had moderate acceptance; cultured meat had lowest acceptance. Indian consumers were more positive toward new food technologies.
4	Chandramenon et al.	2024	Computers and Electronics in Agriculture	Smart Aquaponics 4.0 and water quality monitoring	IoT and AI technologies enable automated monitoring, water quality prediction, and improved productivity in aquaponics systems.

Research

Gap

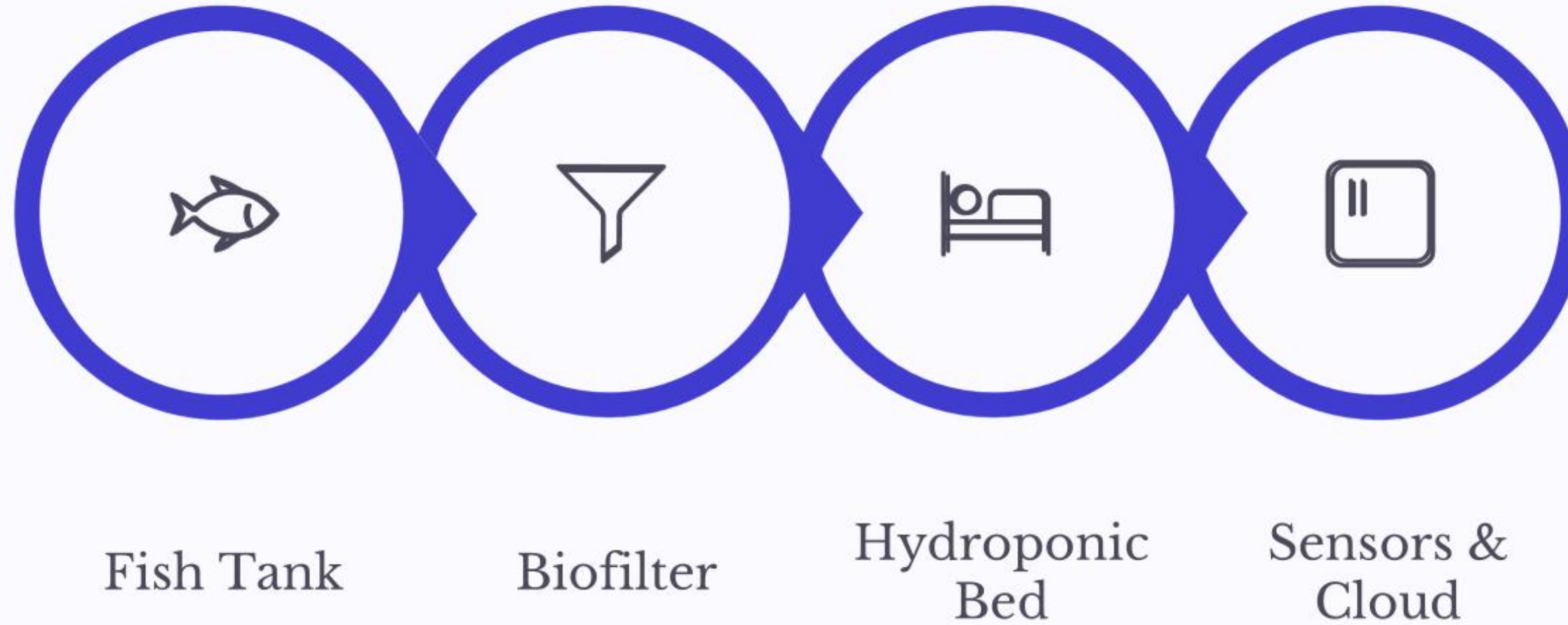
- Limited studies on consumer acceptance in developing countries
- Gap between laboratory-based research results and real-world agricultural implementation
- Insufficient analysis of emotional and behavioral factors influencing adoption
- Lack of long-term performance studies for greenhouse cooling materials
- Limited research on commercial-scale aquaponics profitability and scalability
- Lack of research on integrated sustainable food production systems

Methodology

- Studied total of 8 research papers and selected 4 best papers for our literature review
- Identified the major research gaps from those papers
- Conducted a pilot market survey
- Market research about aquaponics and hydroponics markets
- Developed a sample system architecture
- Created a process flow diagram
- Initial simulation done with the help of zoho IoT platform

System Architecture

The Blueprint for Smart Aquaponics



Our system integrates key components for a seamless and intelligent urban farming experience.

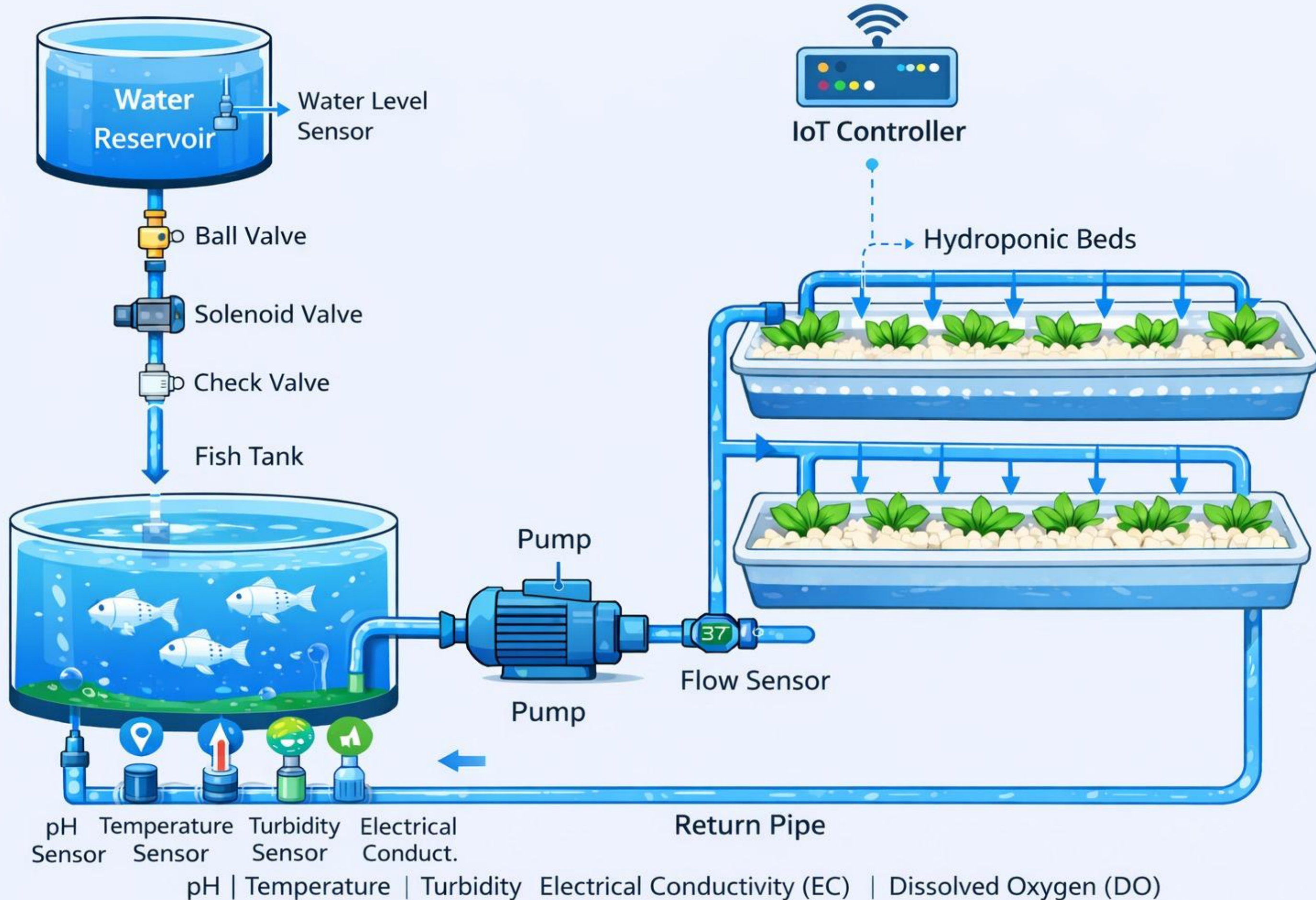
Core Components

- Fish Tank (Aquaculture Unit)
- Hydroponic Grow Beds
- Automated Valves & Pumps
- AI Vision Module
- Cloud Dashboard & Mobile App

IoT Sensor Network

- pH sensor
- Temperature sensor
- Turbidity sensor
- EC/TDS sensor
- Water level sensor

Process flow diagram



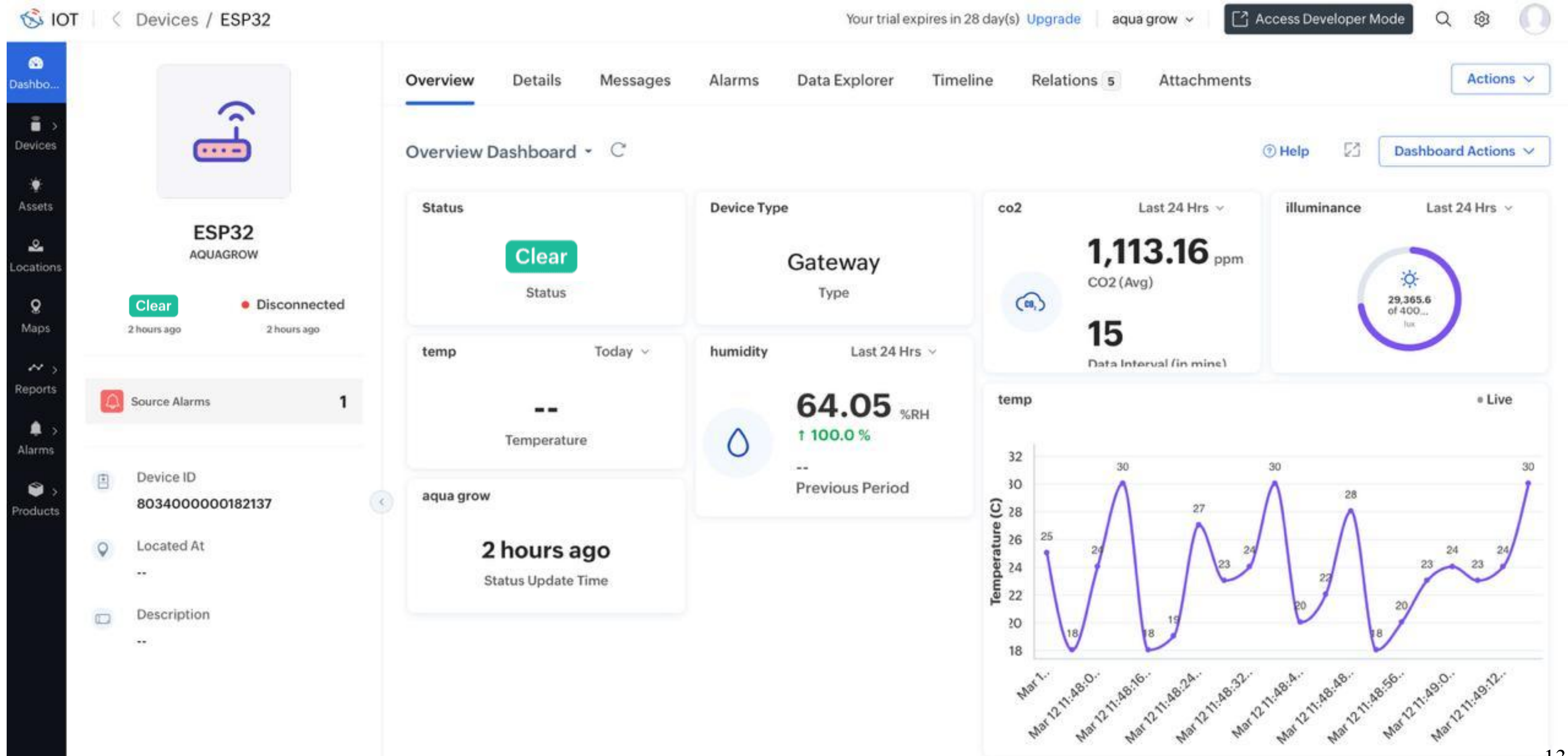
Simulation

- To test our smart hydroponics monitoring system, we used Zoho IoT Device Simulation.
- The ESP32 microcontroller was configured as a virtual IoT device.
- Sensor data was simulated to replicate real greenhouse conditions.
- The platform allows real-time monitoring, visualization, and testing without physical hardware.
- This helps in validating the system logic, thresholds, and automation before actual deployment.

Simulation setup

Icon	Datapoint Name	Data Type	Values		Unit
	Light Level ESP32	Random 	20000 Minimum value	40000 Maximum value	lux
	CO2 ESP32	Random 	800 Minimum value	1500 Maximum value	ppm
	Humidity ESP32	Random 	50 Minimum value	80 Maximum value	%RH
	Temperature ESP32	Random 	18 Minimum value	30 Maximum value	C

Dashboard and Visualization



Market research-Primary data collection

Survey Method

- A structured questionnaire survey was conducted to gather expert opinions on the feasibility and market potential of the AquaGrow system.
- The questionnaire included multiple-choice questions covering feasibility, demand, product advantages, challenges, and adoption potential.

Sampling Method

- Purposive Sampling Method was used.
- Respondents were selected based on their knowledge and experience in agriculture, hydroponics, or smart farming technologies.

Sample Size

- Sample size: 10 respondents
- The respondents were selected from agriculture and smart farming related backgrounds to ensure relevant and informed responses.

Respondents

The survey participants included:

- Agricultural Scientists
- Research Scholars
- Agricultural Engineering Students

From institutions such as:

- Tamil Nadu Agricultural University (TNAU)
- Kerala Agricultural University (KAU)

Key Insights

- Most respondents consider portable IoT-enabled aquaponics systems feasible for urban farming in limited spaces
- Survey results indicate growing demand for smart, sustainable food production technologies in urban areas
- Key advantages identified include water conservation, nutrient recycling, and automated monitoring
- Experts believe IoT and AI integration can significantly improve crop monitoring and system efficiency

Questionnaire

SECTION A: MARKET FEASIBILITY & DEMAND

1. Do you think a portable integrated aquaponics system is feasible for urban farming?

A) Highly feasible B) Moderately feasible C) Slightly feasible D) Not feasible

2. How suitable do you think such a system is for urban spaces like rooftops or balconies?

A) Very suitable B) Suitable C) Slightly suitable D) Not suitable

3. How would you rate the demand for smart farming technologies in urban areas?

A) Very high B) High C) Moderate D) Low

4. Do you think urban households are interested in growing their own vegetables using smart farming systems?

A) Yes B) Maybe C) No

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SECTION B: PRODUCT EVALUATION

5. What do you think are the major advantages of an integrated aquaponics system?

A) Water conservation B) Efficient nutrient recycling C) Higher productivity in small spaces D) Reduced fertilizer usage E) Automated monitoring

6. What challenges might occur when implementing such systems in urban environments?

A) Initial setup cost B) Technical maintenance C) Power supply reliability D) Limited installation space E) Lack of awareness

7. How important are IoT sensors and AI technologies for improving aquaponics systems?

A) Very important B) Important C) Moderately important D) Not important

Questionnaire

SECTION C: MARKET ADOPTION & STAKEHOLDERS



SECTION D: MARKET POTENTIAL

8. Who do you think are the main users of a portable aquaponics system?

- A) Urban households B) Restaurants and cafes
C) Educational institutions D) Rooftop farming communities
E) Agri-tech startups

9. Who are the major stakeholders involved in such systems?

- A) Farmers B) Government agencies
C) Research institutions D) Agri-technology companies
E) Consumers

10. Would you recommend adopting such systems for urban farming?

- A) Yes B) Maybe C) No

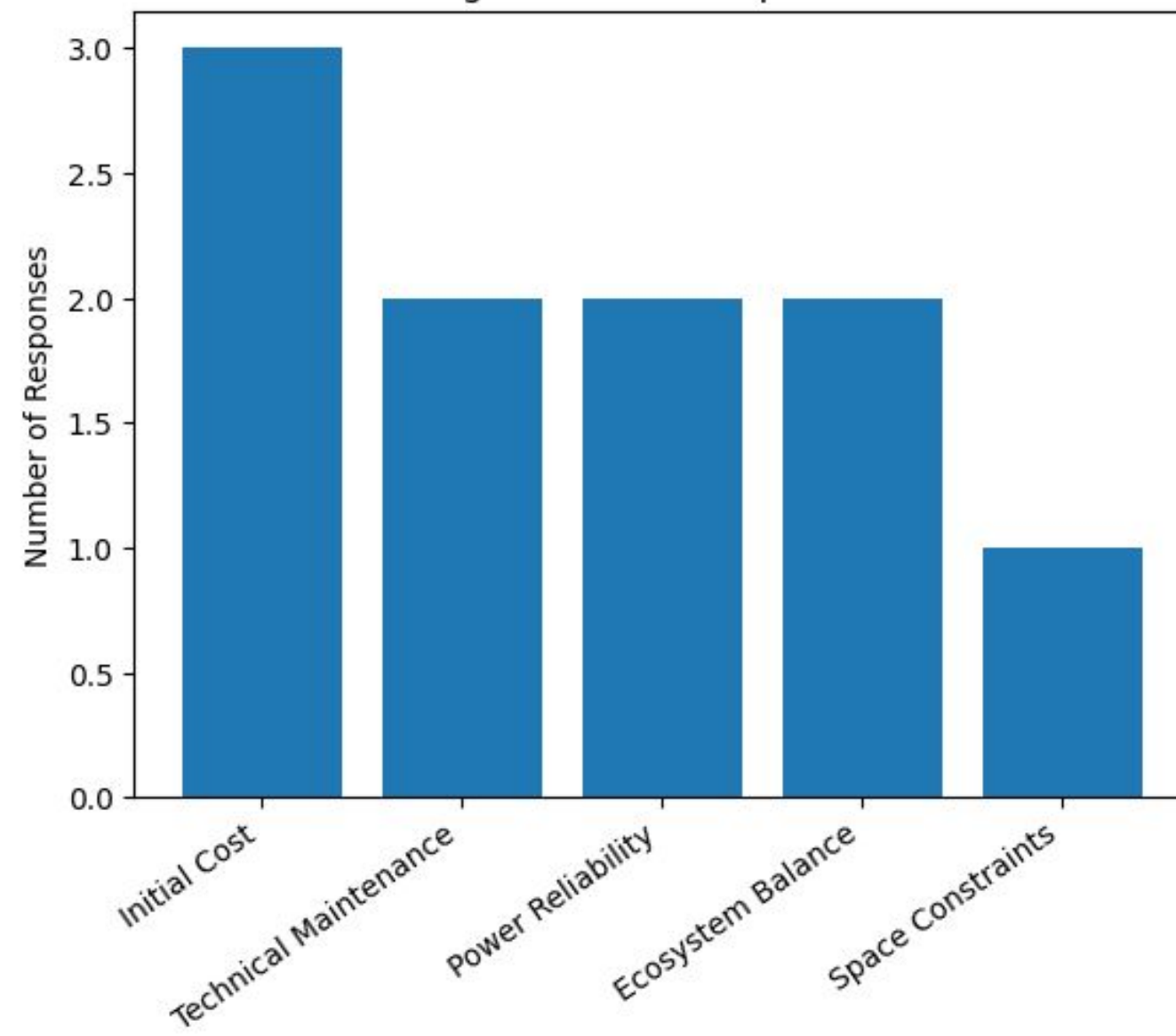
11. What price range would you consider affordable for a compact smart aquaponics system?

- A) ₹10,000–₹20,000 B) ₹20,000–₹40,000
C) ₹40,000–₹60,000 D) Above ₹60,000

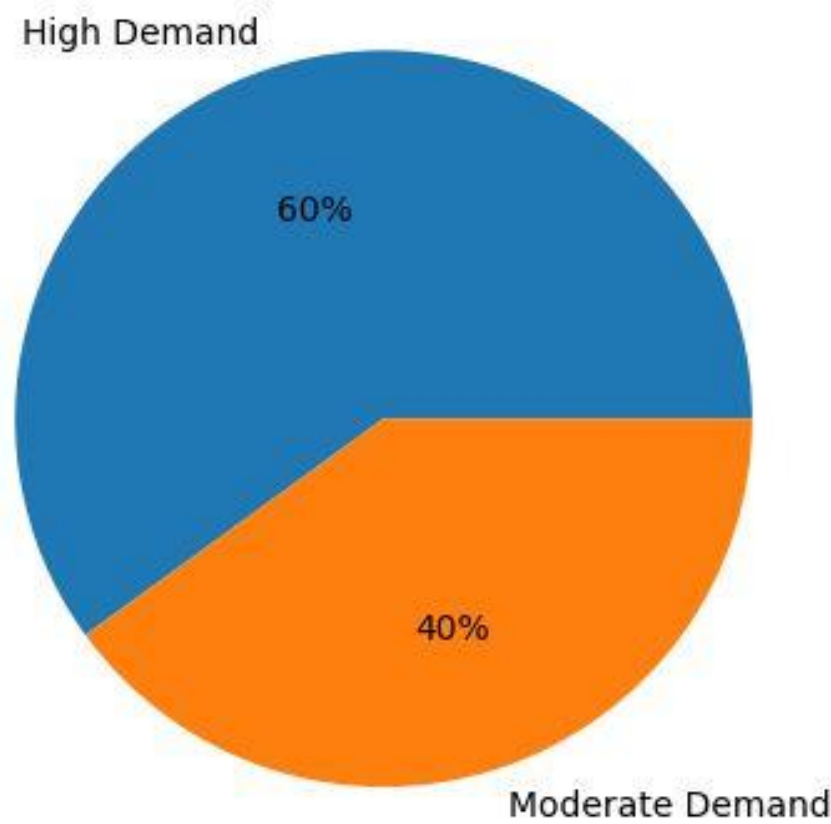
12. What is the future potential of IoT-enabled aquaponics systems in urban agriculture?

- A) Very high potential B) High potential
C) Moderate potential D) Low potential

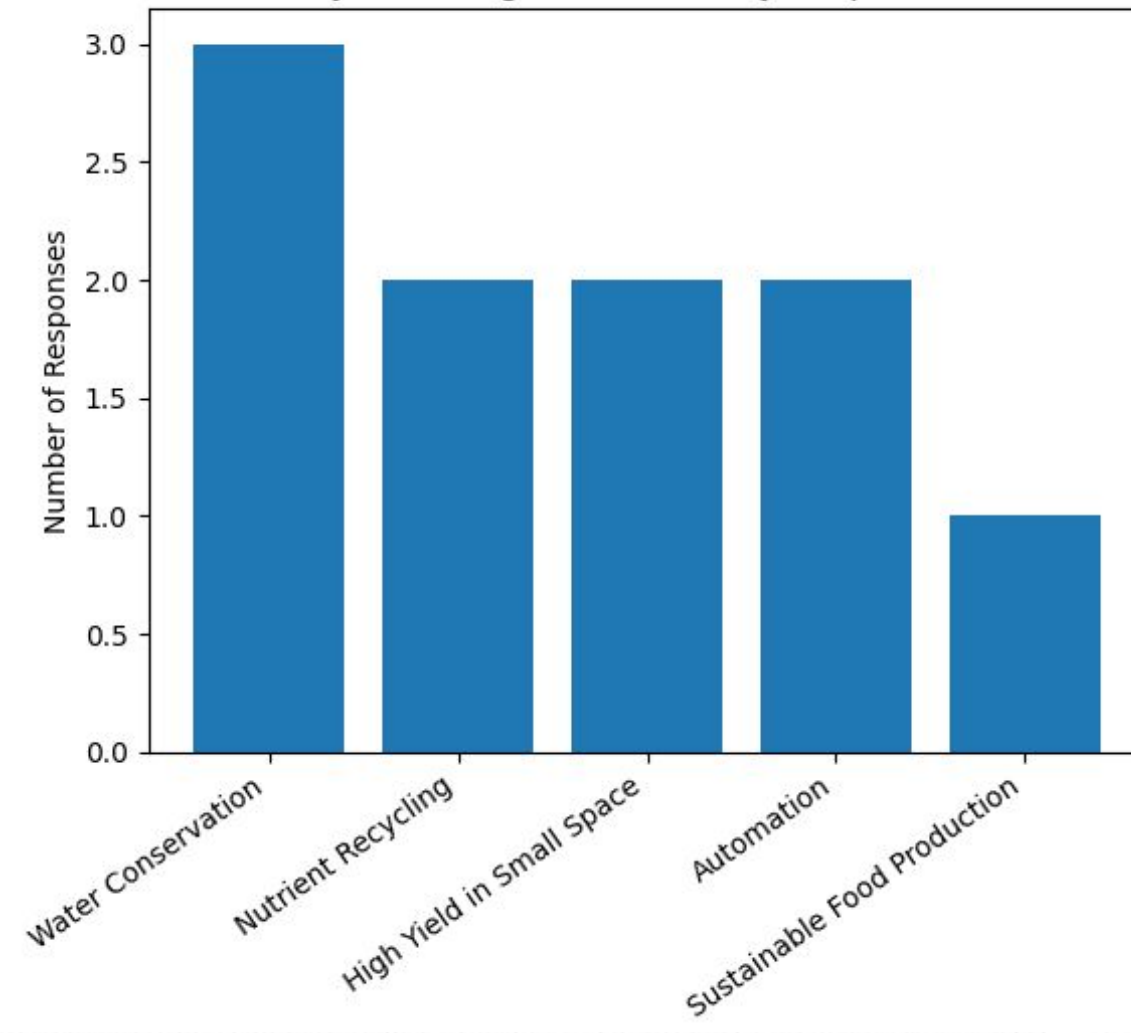
Challenges for Urban Implementation



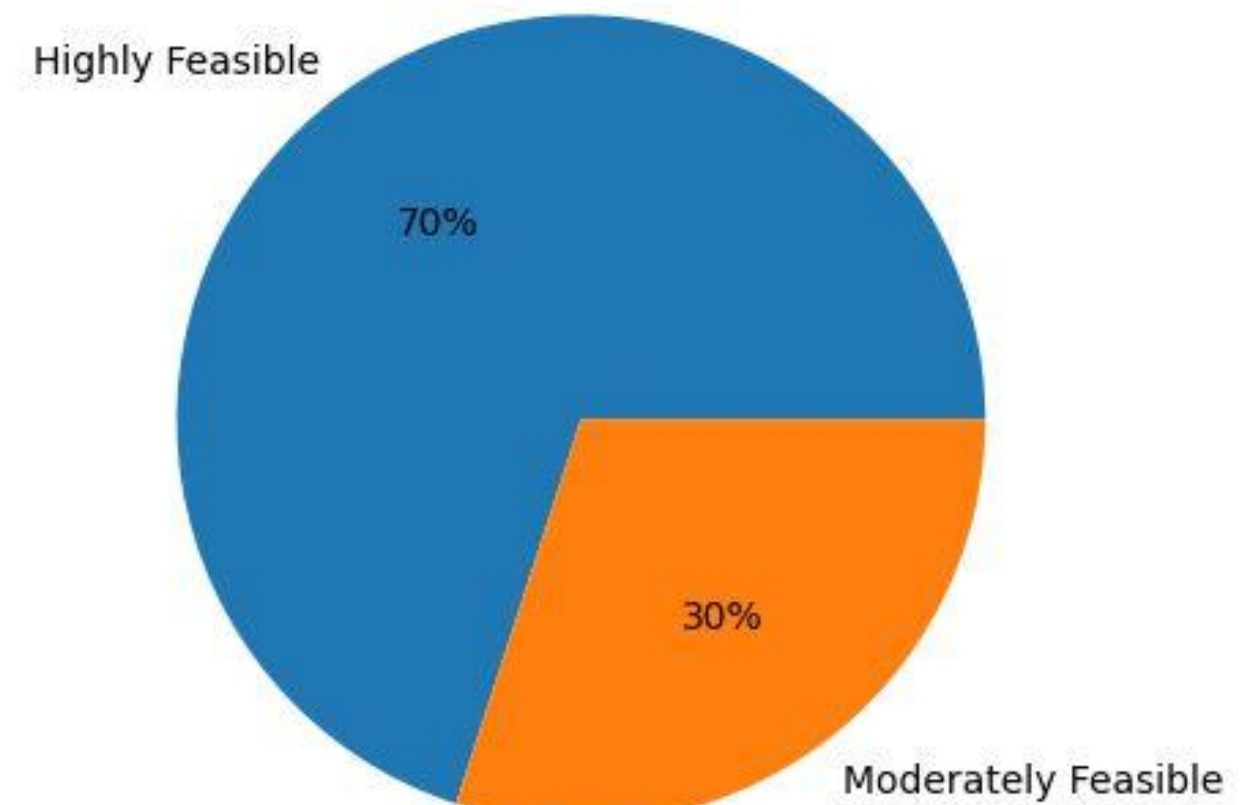
Market Demand for Smart Urban Farming Systems



Key Advantages Identified by Respondents



Feasibility of Portable IoT Aquaponics System (n=10)



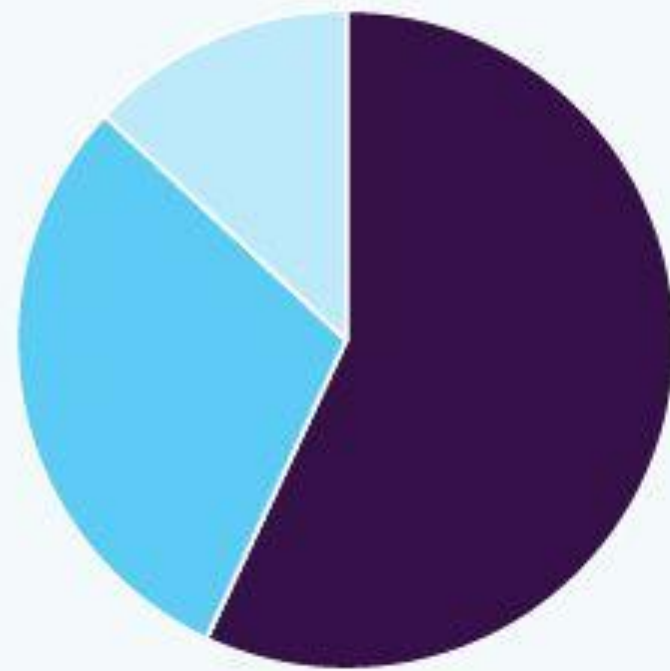
Market Research-Secondary Data Collection

Aquaponics Market

- Market size (2024): USD 1.08 Billion
- Expected market size (2030): USD 2.29 Billion
- CAGR: ~13.5%

Aquaponics Market

Share, by Application, 2024 (%)



● Commercial ● Home Production ● Research & Education

GRAND VIEW RESEARCH

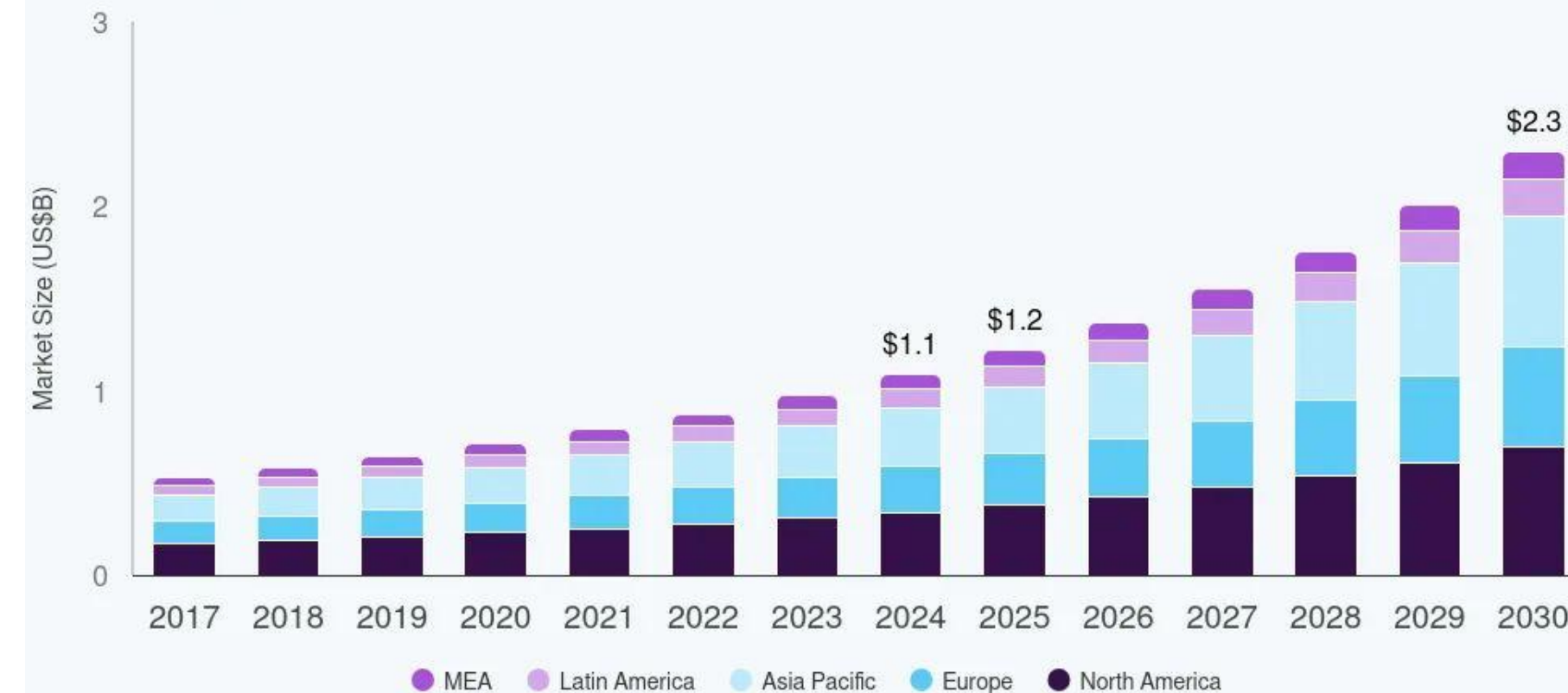
\$1.1B

Global Market Size,
2024

Source:
www.grandviewresearch.com

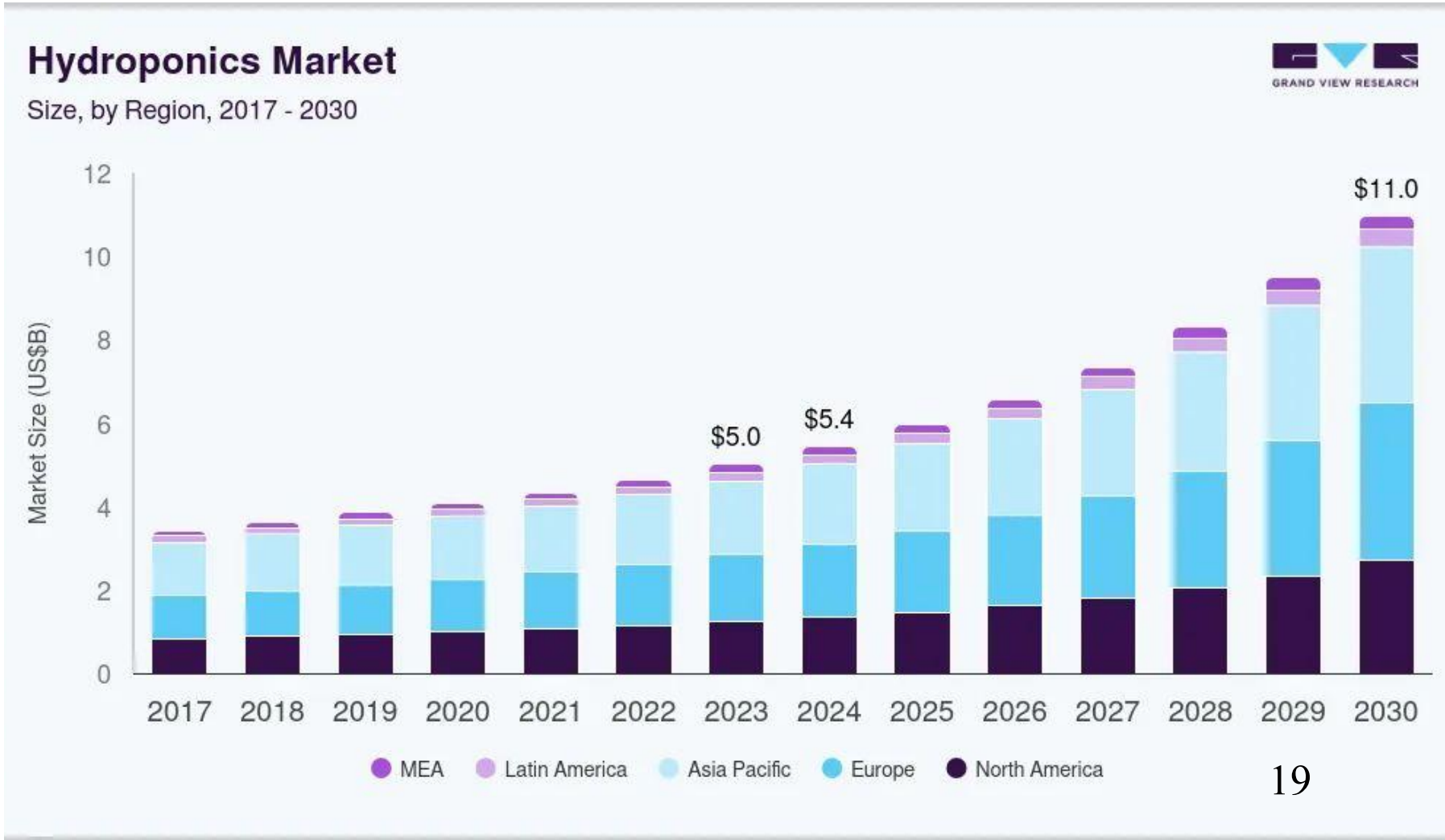
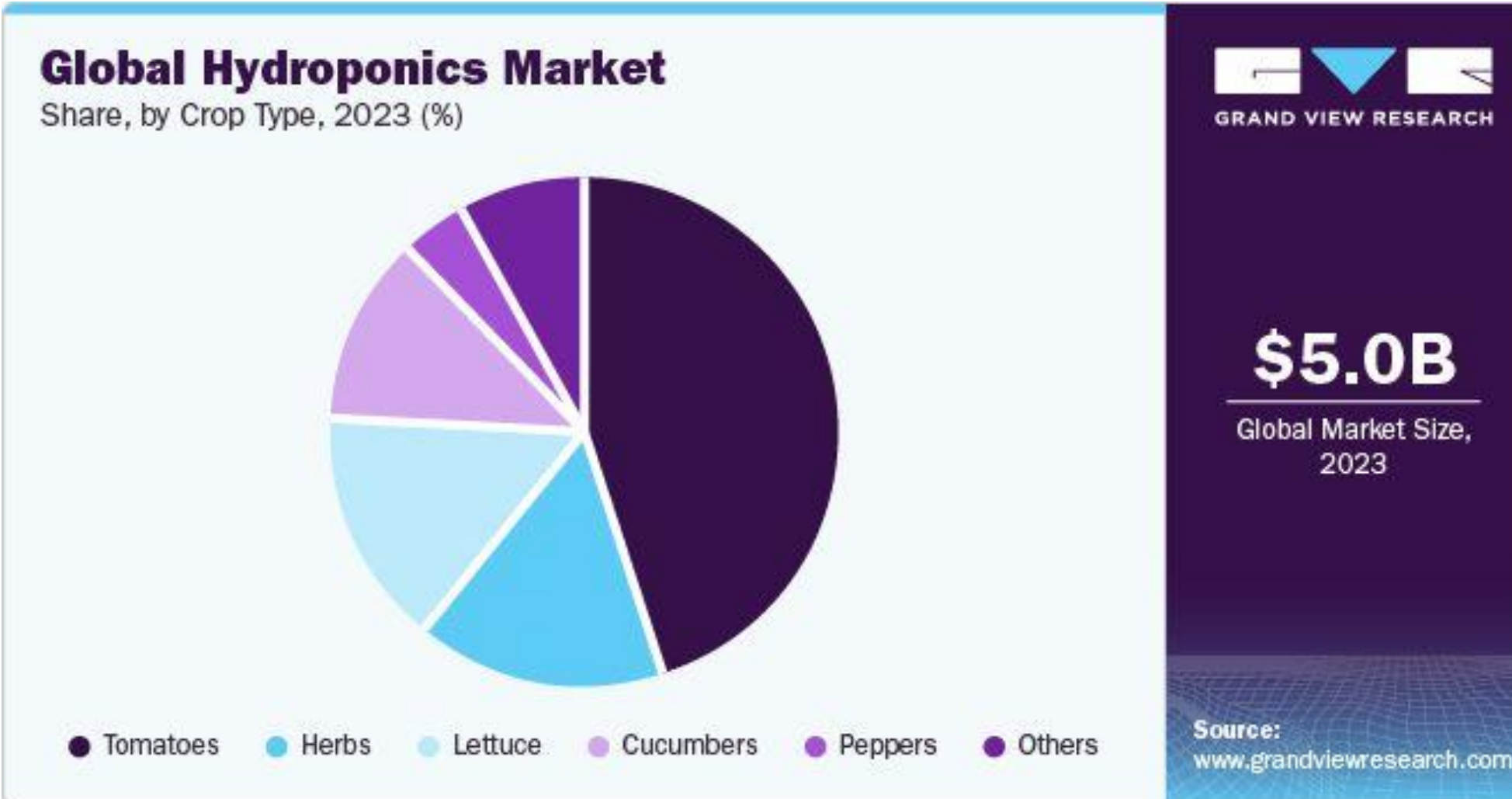
Aquaponics Market

Size, by Region, 2017 - 2030



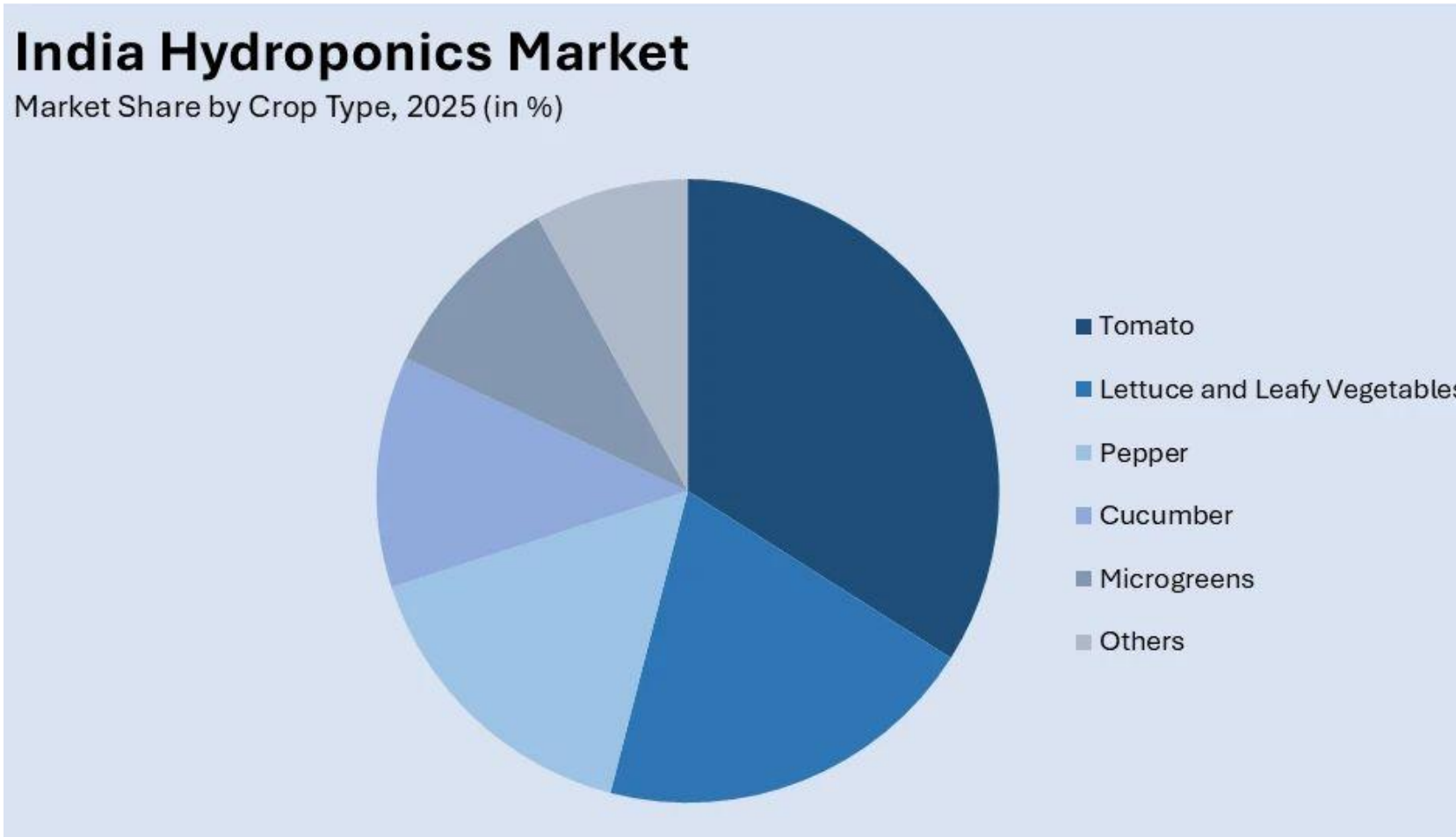
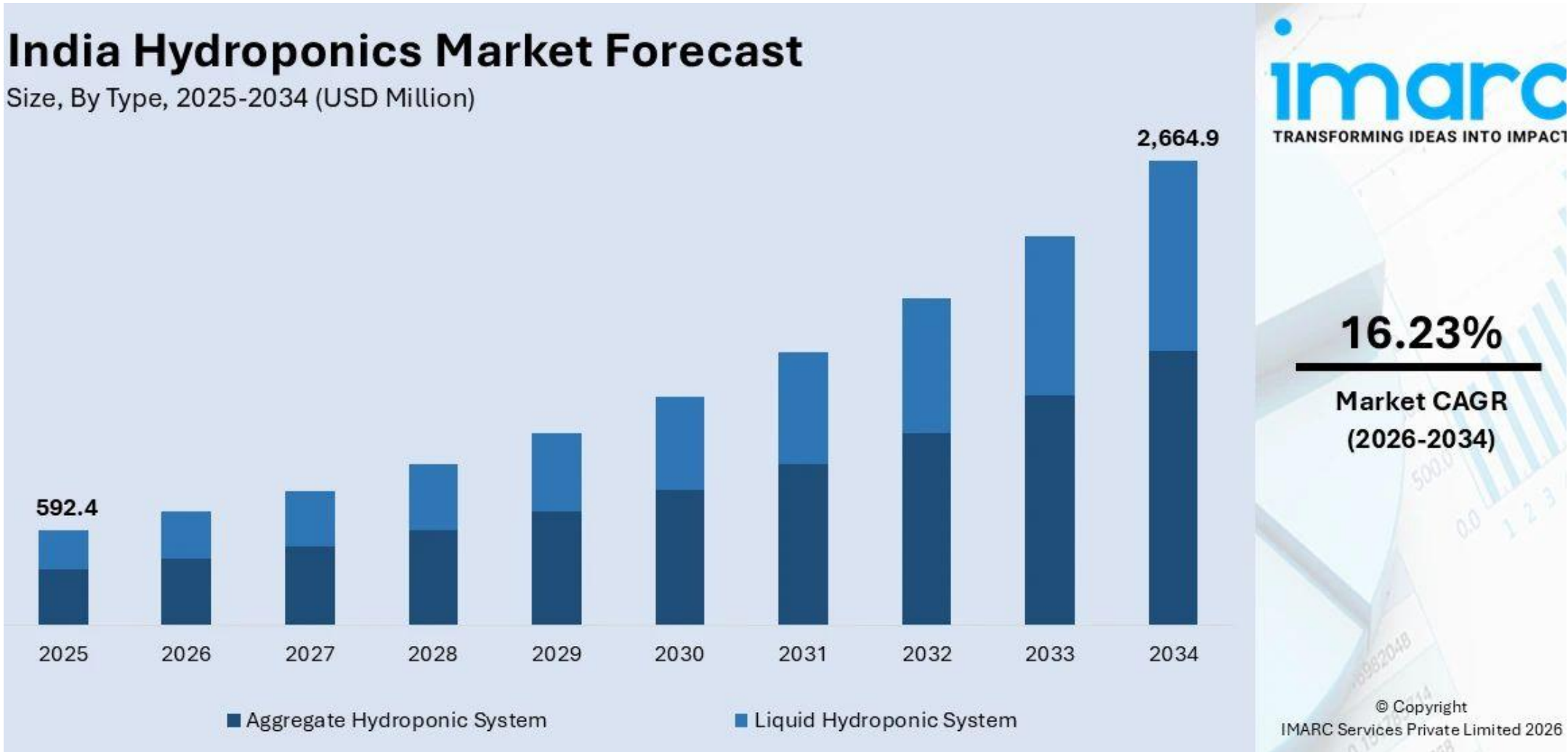
Hydroponics Market

- Market size (2023): USD 5 Billion
- Expected market size (2030): USD 10.98 Billion
- CAGR: ~12.4%



India Hydroponics Market

- Market size (2025): USD 592.4 Million
- Expected market size (2035): USD 2 Billion
- CAGR: ~17 %



Conclusion

- Integration of aquaculture, hydroponics, and IoT sensors enables efficient monitoring of environmental parameters and optimized resource usage.
- Simulation results confirm that real-time monitoring and automation can improve system reliability and crop management.
- Market analysis indicates a rapidly growing global aquaponics and hydroponics industry, highlighting strong commercial potential.
- Survey responses also show that urban users and institutions consider such systems feasible for small-space food production.

Reference

- Giacalone, D., & Jaeger, S. R. (2023). Consumer acceptance of novel sustainable food technologies: A multi-country survey. *Journal of Cleaner Production*, 408, 137119. <https://doi.org/10.1016/j.jclepro.2023.137119>
- Zou, H., Wang, C., Yu, J., Huang, D., Yang, R., & Wang, R. (2023). Eliminating greenhouse heat stress with transparent radiative cooling film. *Cell Reports Physical Science*, 4(8), 101539. <https://doi.org/10.1016/j.xcrp.2023.101539>
- Love, D. C., Fry, J. P., Li, X., Hill, E. S., Genello, L., Semmens, K., & Thompson, R. E. (2015). Commercial aquaponics production and profitability: Findings from an international survey. *Aquaculture*, 435, 67–74. <https://doi.org/10.1016/j.aquaculture.2014.09.023>
- Giacalone, D., & Jaeger, S. R. (2023). Consumer acceptance of novel sustainable food technologies: A multi-country survey. *Journal of Cleaner Production*, 408, 137119.
- <https://www.grandviewresearch.com/industry-analysis/aquaponics-market-report?utm>
- <https://www.grandviewresearch.com/industry-analysis/hydroponics-market>
- <https://www.marketresearchfuture.com/reports/india-hydroponics-market-46718>

Thank You